

MSc. in Computing Practicum Approval Form

Section 1: Student Details

Project Title:	Aspect – Based Sentimental Analysis of Student Feedback Using Large Language Models.
Student ID:	A00053422 , A00053471
Student name:	Kavya Kumar, Sumukha Sagar
Student email	kavya.kumar2@mail.dcu.ie , sumukha.sagar2@mail.dcu.ie
Chosen major:	Data Analytics
Supervisor	Yalemisew Abgaz
Date of Submission	21 – 10 - 2025

Section 2: About your Practicum

What is the topic of your proposed practicum?

This practicum focuses on applying Sentimental Analysis and Aspect – Based Sentimental Analysis (ABSA) to four years of student feedback and discussion data consisting of approximately 4k feedback and discussion entries along with 1k labels. The data is entirely text based. The aim is to classify each comment as positive, negative, or neutral and then extract specific aspects, such as assignments, lectures, or flexibility, that influence the sentiment. We will fine – tune or use and adapt existing Large Language Models(LLMs) to improve domain – specific classification accuracy. The final outcomes will identify common concerns, highlight positive themes, and assist educators in understanding student experiences more deeply.

Details of the papers related to this topic.

1. **Kumar, M. (2025).** *Advancements in Aspect-based Sentiment Analysis: Leveraging Deep Learning Algorithms for Accurate Sentiment Classification.* ICAISS-2025, IEEE.
2. **Varshaa, B., & Prabu, M. (2025).** *Aspect Extraction and Sentiment Analysis on E-commerce Reviews using Deep Learning.* ICCRTEE-2025, IEEE.
3. **Nuha, U., & Lin, C.-H. (2023).** *Aspect-Based Sentiment Analysis with Semi-Supervised Approach on Taiwan Social Distancing App User Reviews.* ICAIIC 2023.
4. **Zhang, W., Li, X., Deng, Y., Bing, L., & Lam, W. (2023).** *A Survey on Aspect-Based Sentiment Analysis: Tasks, Methods, and Challenges.* Singapore Management University, School of Computing and Information Systems.

5. **Zhao, H., Yang, M., Bai, X., Fu, J., et al. (2024).** *A Survey on Multimodal Aspect-Based Sentiment Analysis*. IEEE Access.

How does your proposal relate to existing work on this topic described in these papers?

The selected papers collectively form a strong foundation for understanding sentiment analysis and aspect – based sentiment analysis (ABSA). Kumar (2025) demonstrates that deep learning architectures significantly improve sentiment classification accuracy, supporting our use of LLM – based techniques for student feedback. Varsha and Prabu (2025) show how aspect extraction and sentiment detection can reveal detailed user opinions in real – world applications, which directly relates to our goal of identifying the most frequently mentioned aspects in student comments.

Nuha and Lin(2023) proposed a semi – supervised ABSA approach for unlabelled data, combining lexicon scores with metadata. This aligns closely with our project, which also involves a limited labelled dataset and the need for efficient, scalable annotation methods. Their methodology informs our approach to developing a customized LLM model using partial training data.

Zhang et al. (2023) provide a comprehensive review of ABSA tasks, deep learning methods, datasets, and transformer-based models. This survey validates our choice of using LLMs and highlights challenges such as domain adaptation, which is relevant for student feedback analysis. Finally, Zhao et al. (2024) examine multimodal ABSA, emphasising the trend toward richer, aspect-level understanding. Together, these works justify our methodology and help shape our experimental design and evaluation strategy.

What are the research questions that you will attempt to answer?

This project aims to address several key research questions related to the analysis of student feedback.

1. **How accurately can a customised Large Language Model classify student feedback into positive, negative, and neutral categories?**

This involves evaluating whether fine-tuning improves sentiment detection over zero-shot or pre-trained LLM performance.

2. **Which aspects appear most frequently in student feedback, and how are these aspects associated with particular sentiment categories and change over the year?**

This aspect-level insight will help identify which features—such as assignments,

teaching quality, communication, or workload—positively or negatively influence student experiences. Exploring temporal trends will show whether student perceptions have improved, declined, or remained stable.

Together, these research questions guide the analytical framework for evaluating the usefulness and accuracy of LLM-based sentiment and aspect extraction methods within an educational context.

How will you explore these questions?

- What software and programming environment will you use?

We will use Python with libraries such as Pandas, Matplotlib, PyTorch, scikit – learn, and Jupyter Notebook or VS Code for development.

- What coding/development will you do?

Development will include pre-processing text data, fine – tuning an LLM for sentiment classification, implementing rule – based or model – based aspect extraction, and creating evaluation scripts for performance metrics.

- What data will be used for your investigations?

The dataset includes approximately 4k student feedback entries and student discussion posts also 1k labels collected over four academic years. All data is text – based. The dataset provided contains only anonymised student feedback with all personal identifiers removed. As a result, the data poses no privacy risks and can be used safely for research purposes in accordance with ethical and institutional guidelines.

- Is this data currently available, if not, where will it come from?

Yes, the data is currently available and will be supplied directly by the supervisor.

- What experiments do you expect to run?

Experiments will involve comparing different LLMs, evaluating fine – tuning, testing different aspect extraction strategies, and analysing aspect frequency across sentiment categories.

- What output do you expect to gather?

Outputs include sentiment labels for all comments, extracted aspects, aspect frequency charts, model accuracy metrics, confusion metrics etc.

- How will the results be evaluated?

Results will be evaluated using accuracy, precision, recall, F-1 score, confusion matrices, and qualitative analysis of extracted aspects to assess how well the model captures student sentiment and feedback themes.